25901054 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Skill Development Components (SDC) - MG1SDCSDS102 - MATHEMATICS FOR SYSTEM & SOFTWARE

(2025 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Determine whether i) 3 |7 ii) 3|12. [A] / [CO1]

2. What is the relation connecting, the number of elements in any two sets and the number of elements in their union and intersection?Give an example. [U] / [CO2]

3. Find the adjoint of the matrix A if the cofactor of the matrix A is $\begin{bmatrix} 1 & 8 & 7 \\ 2 & 4 & 1 \\ 6 & 9 & 5 \end{bmatrix}$ [A] / [CO3]

4. Define trace of a square matrix. Give an example. [A] / [CO3]

5. Define Adjacent Vertices with an example [An] / [CO4]

6. Define Graph with an Example [An] / [CO4]

7. Write the three operations in Boolean algebra [An] / [CO5]

8. Write the Distributive laws in Boolean Algebra [An] / [CO5]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

9. Verify $ab \equiv ((a \bmod m)(b \bmod m)) \pmod m$ using a=13, b=7,m=5 [A] / [CO1]

10. State the two De Morgan's Law in set theory. Prove it with membership tables [U] / [CO2]

11. If $A = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$, prove that A adjA = |A| I, where I is the unit matrix of order 2 [A] / [CO3]

12. Define idempotent matix. Check whether $A = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 4 & 1 \\ 2 & 1 & 2 \end{bmatrix}$ is idempotent or not [A] / [CO3]

13. State Handshaking theorem, then find the number of edges in a graph with 10 vertices each of degree six? [An] / [CO4]

14. Draw the following graphs and label the vertices (a) C_5 (b) W_4 (c) K_5 (d) $K_{1,5}$ [An] / [CO4]

15. Find the sum of product expansions for the function $F(x, y, z) = xy + y\bar{z}$ [An] / [CO5]

16. If $F = \bar{x}y + yz$ and $G = \overline{xyz}$, find the values F+G and F.G using tables. [An] / [CO5]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

17. i) Encrypt the message IN THE BOOK using the function f(p) = (2p +5) mod 26 [A] / [CO1]
    ii) Generate 10 Pseudorandom Numbers using $x_{n+1} = (4x_n + 3) \bmod 4$. with $x_0=3$

18. Solve: 2x - y = 0; x + 3y = 7 using i) Cramers rule ii) Matrix inversion method [A] / [CO3]

19. a) Define Euler Circuit and Euler Path explain the criteria to exist Euler Circuit and Euler Path in graph.(b) *Show that K_n has a Hamilton circuit whenever $n \geq 3$* [An] / [CO4]

20. i) Explain AND gate ii) Construct a circuit having the out put $(x + y + z)(\bar{x} + \bar{y} + \bar{z})$ [An] / [CO5]
    iii) Construct a circuit with three varibles having only OR and Inverter

[15x2 = 30]